REFEREE AID - HEAVY WEAPONS & EXPLOSIVES

Heavy Weapons: Heavy Weapons is the collective term for Rocket Launchers, Grenade Launchers, ATGMs, Vehicle Cannons, Mortars, and Howitzers. Grenades are counted as heavy weapons as well, but have some special rules.
This aid is intended to cover the use of heavy weapons against both infantry and vehicles, and will note when to switch to the **Vehicle Combat Referee Aid** when appropriate.

**Machine guns only use STR and HW skill dice when rolling to attack (or AGI and HW when mounted in a vehicle or when using a tripod), but otherwise use the Ranged Combat rules and should not use this aid.*

Combat initiated → Draw initiative cards. Lowest card goes first. → Participant banks 2 fast actions, or 1 fast and one slow action.
- Movement or fast action(s) taken
- Free Actions

→ Aim (fast action) → Determine all (remaining) modifiers to hit. → Ranged Attack - Roll STR and Heavy Weapons dice, plus any ammo dice used

No Quick Shots are available with heavy weapons. Aiming with most heavy weapons is a fast action.
Aiming with *Howitzers* and *Mortars* is a slow action.

To-Hit Variables:
When throwing **grenades**, roll AGI and Mobility instead of STR and HW. For firing weapons on a **tripod or vehicle mount**, use AGI and HW instead of STR and HW.
When firing **artillery (mortars or howitzers)**, roll INT and HW instead of STR and HW.

Grenades: Grenades use an AGI and Mobility roll to hit. Range is always short. Pulling the pin is a fast action, and no aiming is necessary. If a grenade lands in your hex, you can roll a mobility check to drop prone (free action). pg. 70.

To Hit Modifier Ex. pg. 65:

Range S/M/L/E	-/-1/-2/-3
Player Specialties	+1
Scope (slow aim)	+1/+2
Target in Same Hex	-1/-2
Target Hex Terrain	-1/-2
Target Prone (no same hex)	-1
Called Shot	-2
Target moving	-1
Firing from moving vehicle	-2
Firing from elevation	+1
Light/Dim/Dark	-/-1/-2
Rain/Wind/Snow	-1
Dense Smoke	-3
Target in Full Cover	-3
Large Target	+2
Small Target	-2
Defenseless (Same Hex)	+3

Targeting:
Heavy weapons are often used to attack larger targets, such as hexes, vehicles, etc. Damage from heavy weapons can cause both direct damage *and* blast damage from the related explosion Aiming at smaller targets with heavy weapons (even grenades), such as an individual, nets a -2 to-hit modifier.

Targeting a hex: Do not add the Target Hex Terrain modifier when targeting a hex for an attack. Extra ◌ beyond the initial success on the attack roll have no effect.

Hit? → Target Missed (End of participant's turn)

Hit? → Push? → Player may reroll ◌ at their own discretion. → Reroll any STR and Heavy Weapons dice that aren't ✸s or ◌s.

Hit? → Deviation* → Target Missed (End of participant's turn)

**Please note that you can miss with an explosive shot, and yet still have the target damaged by the resultant blast.*

Deviation:
If an attack with an explosive effect misses, roll 2D6 to see where the shot lands. The first D6 determines the direction of the deviation. The second D6 determines how many hexes away the explosion deviates by (cannot exceed more than half the total distance to the target).

Corrections: If an attack from a **howitzer or mortar** misses its target hex, the firer can correct with subsequent shots. Corrections count as a slow action, gives a +1 bonus if the next attack is aimed at the same target. This effect can stack for up to +3 from corrections.

Indirect Fire: Artillery, Mortars, and Grenade Launchers are capable of indirect fire. To hit a target with indirect fire, you need a spotter who can see the target and communicate with you. Directing indirect fire is a Recon roll (slow action), and needs to be done before the weapon is aimed. If the spotter's roll fails, the target can still be fired upon, but the shot will miss (shot will deviate).
If the spotter's roll succeeds, the attack will be rolled as normal; extra spotter roll success give no bonus. Spotters can call in corrections with a slow action (no roll needed) for howitzers and mortars, but the firer will need to make the corrections (slow action).

Hit? → Direct Damage? → Direct Damage: Weapon Base DMG. Add +1 for each additional ◌

Weapon jams can be cleared with a successful Ranged Combat or Heavy Weapons roll. pg. 66.
MISHAP: A single ✸: -1 Rel from weapon. Two or more ✸s: Weapon jam

Damage: If an explosive round/grenade strikes it's target, note that it will do both direct damage **and** blast damage. If the round simply lands nearby however, or is targeting the hex, it will only do blast damage.

Direct Damage? → Blast Damage calculated and target rolls CUF

This is where you can switch to the Vehicle Combat Chart if your target is a vehicle. Remember to determine Penetration!

Blast Damage calculated and target rolls CUF → Target is a vehicle? → *Continue to "Check angle of attack" on the Vehicle Combat Referee Aid*

Target is a vehicle? → Determine Hit Location by rolling 1D6

Hit Location

1	Legs
2-4	Torso
5	Arm
6	Head

Cover/Armor Ex.

Concrete Wall	3
Brick Wall	2
House Wall	2
Indoor Wall	1
Sandbags	4
Barbed Wire	-
Flak armor	1
Steel helmet	1
Plate armor	2
Kevlar helmet	1

Determine Hit Location by rolling 1D6 → Cover/Armor? → Apply weapon's armor modifier if there is one.

If armor level is 2 steps higher than the base weapon damage, the attack is fully deflected. pg. 73

Cover:
Partial Cover: Covers all body parts except arms and head.
Full cover: Covers all body parts. Targets can still be fired on full cover with a -3 modifier.

EXPLOSIONS

BLAST POWER	DICE	DAMAGE	CRIT	ARMOR
A	2xD12	4	3	+1
B	2xD10	3	3	+1
C	2xD8	2	3	+1
D	2xD6	1	2	+1

Apply weapon's armor modifier if there is one. → Reduce damage by armor value.

Apply armor ablation to armor or cover. Roll 1D6. -1 Rel for each ✸ rolled. pg. 73.

Reduce damage by armor value. → Greater than critical threshold?

If remaining dmg = the weapon's critical threshold +2, roll twice on the crit table. pg. 74.

Greater than critical threshold? → Roll on the critical tables pg. 76 → Apply damage and any criticals. Target also suffers 1 stress on a failed CUF roll.

Information on recovering from damage and criticals can be found starting on pg. 73.

Explosions:
Blast Power: Explosive weapons have a blast radius rating from A to D, where A is the most powerful explosion. Each hex away from the strike reduces the blast power by 1 step. Reducing the blast power below D results in the explosion having no effect.
Blast Damage: Roll two of the dice corresponding to the rating of the blast power for each person in the blast radius. Any ◌ results in that person being hit by the blast. Each additional ◌ beyond the first adds +1 damage.
Suppression: Anyone in the blast radius is immediately knocked prone and must make a CUF roll or be suppressed. This does not apply to people in vehicles or behind cover which the blast fails to penetrate.
Prone: If the target is prone, the blast power is reduced one step.
Solid cover: Cover provides protection against blasts just like it does normal fire, unless the blast is in the same hex as the target.

Landmines:
Placing landmines in a hex requires a stretch of work and a tech roll. Success turns the hex into a minefield. Failure still turns the hex into a minefield, but it will be automatically spotted by anyone moving into the hex. You can try rolling again the next stretch to better hide the mines.
Anti-Personnel (AP) mines: These mines explode when stepped on. Detection of a minefield is a passive (no push) Recon roll. Failure means the person moves into the minefield and risks an explosion. The chance of explosion is a D6 - if the result is equal to or less than the number of mines in the hex, an explosion occurs. Moving through a minefield means movement at 1 hex per turn, with a Recon roll required each turn. Failure risks an explosion as above. If detonated, the direct damage from a mine is done to the target's legs. It also sets off a blast which can further damage anyone in the hex. If single file, a group can follow the steps of the leader who would make the necessary rolls.
Anti-Tank (AT) mines: Larger mines that can only be triggered by vehicles (only one per hex). Spotting AT mines is easier (+2 modifier), but that difficulty is cancelled by trying to make a recon roll from a vehicle (-2 modifier). Driving unaware into an AT mined hex automatically sets off the mine. Driving through a minefield requires slow driving (1 hex per round) and a successful driving roll for each hex. Setting off a mine causes direct damage to the vehicle using the vehicle's rear armor level. The mine also triggers a blast in the hex.
Direction mines: Direction mines are set up to be triggered remotely (usually by tripwire). They do only blast damage (no direct damage). Blast power is decreased 1 step for every 3 hexes (see diagram).
Improvised Explosive Device (IED): IEDs can be created using explosives, spare parts (varies according to detonation device), a successful Tech roll, and a shift of work. A failed roll will detonate the device. Additional IED-specific
rules can be found on pg. 70.
Clearing mines: Clearing a hex of mines requires a stretch of work, a knife or stick, and a successful Tech roll. Failure detonates a mine. More advanced gear can offer positive modifiers.

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